

SOME EXAMPLES AND COUNTEREXAMPLES IN OKA THEORY

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ABSTRACT. This paper gives examples and counterexamples in Oka theory: two about deleting closed sets from complex Euclidean space, one about blowing up along a connected Oka center, and one about deforming continuous maps to regular maps. Two positive results show that $\mathbb{C}^3 \setminus S$ is Oka for every closed set $S \subset \mathbb{R}^3$, and that the complement of the closed Hartogs triangle in $\mathbb{C}^2$ is Oka. In contrast, for every $n \geq 3$ there is a proper holomorphic embedding $\mathbb{C} \hookrightarrow \mathbb{C}^n$ whose image A is a closed connected smooth curve biholomorphic to $\mathbb{C}$, but whose blow-up $\text{Bl}_A \mathbb{C}^n$ is Brody volume hyperbolic and hence not Oka. Finally, for every $n \geq 2$, there exist a smooth connected affine algebraic variety X and a continuous map $X \rightarrow \mathbb{C}^n \setminus \{0\}$ that is not homotopic to any regular map $X \rightarrow \mathbb{C}^n \setminus \{0\}$; equivalently, $\mathbb{C}^n \setminus \{0\}$ fails the algebraic basic Oka property (aBOP).

1. INTRODUCTION

Let Y be a complex manifold. A map $f: K \rightarrow Y$ from a compact set $K \subset \mathbb{C}^m$ is called holomorphic if it extends holomorphically to a neighborhood of K . The Oka property will be formulated through the convex approximation property.

Definition 1.1. [10] The manifold Y is *Oka* if, for every integer $m \geq 1$, every nonempty compact convex set $K \subset \mathbb{C}^m$, and every map $f: K \rightarrow Y$ holomorphic on K , there are entire maps $F_j: \mathbb{C}^m \rightarrow Y$ converging uniformly to f on K .

For a systematic account of Oka manifolds, see [14, Chapter 5]. Uniform convergence on K is independent of the choice of a distance inducing the topology of Y , and the Oka property is invariant under biholomorphisms.

This paper gives examples and counterexamples in Oka theory. Two concern complements of specified closed sets in complex Euclidean space; the third concerns the standard analytic blow-up of a complex manifold along a connected closed Oka submanifold; the fourth concerns replacing a continuous map from a smooth affine algebraic variety to a space of the form $\mathbb{C}^n \setminus \{0\}$, where $n \geq 2$, by a regular map in its homotopy class.

Closed subsets of $\mathbb{R}^3$. The closed-set form of Kusakabe's complement theorem used below is formulated in terms of polynomial convexity, so we first fix the relevant convention.

Definition 1.2. For a compact set $K \subset \mathbb{C}^n$, its polynomial hull is

$$\widehat{K} = \left\{ z \in \mathbb{C}^n : |P(z)| \leq \sup_{x \in K} |P(x)| \text{ for every } P \in \mathbb{C}[z_1, \dots, z_n] \right\}.$$

By convention, $\widehat{\emptyset} = \emptyset$, and K is called *polynomially convex* when $\widehat{K} = K$. For a closed set $E \subset \mathbb{C}^n$, a *compact exhaustion* is an increasing sequence $\{E_j\}_{j \geq 1}$ of compact subsets whose union is E and such that every compact subset of E is contained in some E_j . In this setting, set

$$\widehat{E} = \bigcup_{j \geq 1} \widehat{E}_j,$$

and call E polynomially convex when $\widehat{E} = E$.

This is the convention in Kusakabe's theorem used below. It is independent of the exhaustion: if $\{E_j\}$ and $\{F_j\}$ are two compact exhaustions, then every E_j is contained in some F_k , whence $\widehat{E}_j \subset \widehat{F}_k$, and the reverse inclusion follows after interchanging the two exhaustions.

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Kusakabe proved that certain complements of closed polynomially convex sets are Oka and applied this theorem to closed subsets of the standard real subspaces $\mathbb{R}^k \subset \mathbb{C}^n$. His result [21, Theorem 1.6 and Corollary 1.7] covered all pairs (n, k) except

$$(n, k) = (2, 1), \quad (2, 2), \quad (3, 3).$$

Forstnerič and Fornæss Wold settled $(2, 1)$ [17, Proposition 4.9] and recorded that $(2, 2)$ and $(3, 3)$ remained open [17, Remark 4.10]. We settle the case $(3, 3)$.

Theorem 1.3. *If $S \subset \mathbb{R}^3$ is closed in the relative topology, then $\mathbb{C}^3 \setminus S$ is an Oka manifold.*

The missing case $(3, 3)$ is not caused by a failure of polynomial convexity: every compact subset of $\mathbb{R}^m \subset \mathbb{C}^m$ is polynomially convex [9, Example 1]. The difficulty is to meet the geometric hypothesis in Kusakabe's theorem. In the present dimension, that hypothesis requires a holomorphic change of coordinates and a splitting $\mathbb{C}^3 = \mathbb{C} \times \mathbb{C}^2$ in which the Euclidean norm of the second coordinate is bounded on the closed set by a constant times one plus the modulus of the first coordinate. A quadratic polynomial automorphism records the squared length of the last two real coordinates in the imaginary part of the first coordinate and thereby gives this estimate. The same automorphism works for every relatively closed $S \subset \mathbb{R}^3$, so the theorem follows from [21, Theorem 1.6] without a new complement criterion. The remaining exceptional pair $(2, 2)$ is not addressed here.

The closed Hartogs triangle. Write $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$. Forstnerič asked in [12, Problem 4.24] whether the complement of the closed Hartogs triangle

$$H = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1| \leq |z_2| \leq 1\}$$

is Oka. This question was again listed as open in the subsequent survey [13, p. 664]. An affirmative answer to Problem 4.24 follows.

Theorem 1.4. *The complement $\mathbb{C}^2 \setminus H$ of the closed Hartogs triangle is an Oka manifold.*

Forstnerič and Fornæss Wold proved that $(\mathbb{C} \times \mathbb{C}^*) \setminus K$ is Oka whenever $K \subset \mathbb{C}^2$ is compact and polynomially convex [17, Corollary 3.2]. The difficulty is to relate the Hartogs complement to this model. After a complex line through the origin is removed, the ratio of two linear coordinates records the complex direction of a point. For two suitable lines, these directional coordinates identify the corresponding portions of $\mathbb{C}^2 \setminus H$ with complements of the required form. The two portions are Zariski open and cover the whole complement. Kusakabe's localization theorem states that a connected complex manifold is Oka if every point has a Zariski-open Oka neighborhood [20, Theorem 1.4], so it proves Theorem 1.4. Thus the Hartogs inequalities are handled by two explicit local models.

Blow-ups along connected Oka centers. The standard analytic blow-up of a complex manifold X along a closed complex submanifold A replaces each point of A by the projective space of complex lines in its normal space. Denote the blow-up by $\text{Bl}_A X$, its blow-down map by $\pi: \text{Bl}_A X \rightarrow X$, and its exceptional divisor by $\pi^{-1}(A)$. Forstnerič asked which operations, including blow-ups and blow-downs, preserve the Oka property in the list of open problems from his 2011 Krems lecture [11, p. 143]. On the positive side, Lárusson and Truong proved that, for $n \geq 2$, the blow-up of $\mathbb{C}^n$ along any algebraic submanifold (not necessarily connected) is Oka [22, Theorem 1]. On the negative side, Kusakabe used discrete sets constructed by Rosay and Rudin to obtain non-Oka blow-ups of affine spaces along suitable closed discrete centers [20, Lemmas A.1–A.2 and Example A.3]. Our result allows the center to be connected, smooth, and Oka.

For an open set $U \subset \mathbb{C}^n$ and a holomorphic map $F = (F_1, \dots, F_n): U \rightarrow \mathbb{C}^n$, set

$$\text{Jac } F = \det \left(\frac{\partial F_i}{\partial z_j} \right)_{i,j=1}^n, \quad \text{Crit}(F) = \{z \in U : \text{Jac } F(z) = 0\}.$$

The map is *nondegenerate* when $\text{Jac } F \neq 0$, and $z \in U$ is a *regular point* of F when $z \notin \text{Crit}(F)$. More generally, if Y is an n -dimensional complex manifold, a holomorphic map $f: \mathbb{C}^n \rightarrow Y$ is nondegenerate when df has rank n at some point.

Definition 1.5. An n -dimensional complex manifold Y is *Brody volume hyperbolic* if there is no nondegenerate holomorphic map $\mathbb{C}^n \rightarrow Y$.

Here a continuous map $p: X \rightarrow Z$ is *proper* if $p^{-1}(K)$ is compact for every compact set $K \subset Z$.

Theorem 1.6. *For every integer $n \geq 3$, there is a proper holomorphic embedding $\gamma: \mathbb{C} \hookrightarrow \mathbb{C}^n$ whose image $A = \gamma(\mathbb{C})$ is a closed connected smooth complex curve such that $\text{Bl}_A \mathbb{C}^n$ is Brody volume hyperbolic and hence is not Oka.*

The discrete-center examples of Kusakabe use closed discrete sets constructed by Rosay and Rudin that every nondegenerate entire map $\mathbb{C}^n \rightarrow \mathbb{C}^n$ must meet. A connected center creates an additional difficulty. After such a set D is placed on a complex curve, the mere condition $F^{-1}(D) \neq \emptyset$ gives no contradiction in the blow-up argument: one needs an intersection outside $\text{Crit}(F)$. We therefore refine the construction in [25, Lemmas 4.3–4.4 and Theorem 4.5] to obtain a closed discrete set $D \subset \mathbb{C}^n$ such that every nondegenerate entire map $F: \mathbb{C}^n \rightarrow \mathbb{C}^n$ satisfies

$$F^{-1}(D) \not\subset \text{Crit}(F). \quad (1.1)$$

In the present construction the finite subsets on spheres are chosen so that a forced intersection has a regular preimage. Taking their union over successively larger spheres retains this stronger conclusion and produces a closed discrete set.

An interpolation theorem of Forstnerič, Globevnik, and Rosay [16, Proposition 2] places D on a properly embedded copy $A = \gamma(\mathbb{C}) \subset \mathbb{C}^n$. A nondegenerate map to the blow-up would, after composition with the blow-down map, contradict (1.1), because the differential of the blow-down map loses rank along the exceptional divisor. Compared with the earlier discrete-center counterexamples, the center here is connected, smooth, and biholomorphic to the Oka manifold $\mathbb{C}$. It is noncompact, so the theorem does not settle blow-ups at a point or along a compact center.

The algebraic basic Oka property. Here an algebraic manifold is a smooth connected complex algebraic variety, whereas an affine algebraic variety used as a source need not be smooth or connected. A regular map is a morphism of algebraic varieties.

Definition 1.7. [23, pp. 200–201] An algebraic manifold Y has the *algebraic basic Oka property* (aBOP) if, for every affine algebraic variety X , every continuous map $f: X \rightarrow Y$ is homotopic, in the Euclidean topology of complex points, to a regular map $g: X \rightarrow Y$.

The formulation in [12, Definition 6.14(a)] restricts the source to an affine algebraic manifold, hence to a smooth connected source. The source constructed below is smooth and connected, so Theorem 1.8 disproves aBOP under either source convention.

Lárusson and Truong proved that a positive-dimensional algebraic manifold fails aBOP if it is compact or admits a nonconstant regular map from $\mathbb{P}^1$ [23, Theorem 2]. Neither hypothesis holds for $\mathbb{C}^n \setminus \{0\}$: this variety is noncompact, and, for every regular map $f: \mathbb{P}^1 \rightarrow \mathbb{C}^n \setminus \{0\}$, each coordinate of its composite with the inclusion $\mathbb{C}^n \setminus \{0\} \hookrightarrow \mathbb{C}^n$ is a global regular function on $\mathbb{P}^1$ and hence is constant; therefore f is constant. Forstnerič singled out $\mathbb{C}^2 \setminus \{0\}$ in [12, Problem 6.17]; his later survey continued to single out $\mathbb{C}^2 \setminus \{0\}$ as a basic test case for aBOP [13, p. 668].

Theorem 1.8. *For every integer $n \geq 2$, the punctured affine space $\mathbb{C}^n \setminus \{0\}$ does not satisfy aBOP.*

The one-dimensional target $\mathbb{C}^*$ was already known not to satisfy aBOP [7, Theorem 4.2.3]; the case $n = 2$ of Theorem 1.8 answers Problem 6.17.

The construction following [12, Problem 6.17] already gives a regular map from a smooth affine quadric that represents a generator of $\pi_3(\mathbb{C}^2 \setminus \{0\})$. Thus nontrivial target homotopy alone cannot obstruct aBOP; the source must also retain algebraic information. We use the explicit smooth affine variety

$$X_n = \{(x, y) \in \mathbb{C}^2 : y^2 = x^3 - x\} \times (\mathbb{C}^*)^{2n-2}.$$

Throughout the paper, $H^\bullet(-; \mathbb{Q})$ denotes singular cohomology with rational coefficients. The topology of X_n gives a continuous map $F_n: X_n \rightarrow \mathbb{C}^n \setminus \{0\}$ for which

$$F_n^*: H^{2n-1}(\mathbb{C}^n \setminus \{0\}; \mathbb{Q}) \longrightarrow H^{2n-1}(X_n; \mathbb{Q}) \quad (1.2)$$

is nonzero. Deligne associates to the rational cohomology of every complex algebraic variety an increasing filtration by subspaces, called the weight filtration, and regular pullbacks preserve this filtration [4, Theorem 3.2.5(iii)]. Set $Y_n = \mathbb{C}^n \setminus \{0\}$. The calculation in Proposition 5.2 shows that this preservation forces every regular map $X_n \rightarrow Y_n$ to have zero pullback in degree $2n - 1$. This is incompatible with (1.2), since homotopic maps induce the same map in cohomology. This obstruction applies to $\mathbb{C}^n \setminus \{0\}$ for every $n \geq 2$.

Structure of the paper. In [Section 2](#), the polynomial convexity required at the point of use is verified and the quadratic polynomial automorphism is constructed; the section then applies Kusakabe's theorem to closed subsets of $\mathbb{R}^3$. [Section 3](#) contains an explicit coordinate description of the Hartogs complement, the compactness and polynomial convexity arguments, and the application of Kusakabe's localization theorem. [Section 4](#) is devoted to the closed discrete set that every nondegenerate entire map meets at a regular point, its embedding in a connected Oka center, and the proof of the blow-up theorem. [Section 5](#) constructs the source varieties and continuous maps and proves the obstruction to aBOP coming from the weight filtration.

2. COMPLEMENTS OF CLOSED SUBSETS OF $\mathbb{R}^3$

The starting point is the polynomial automorphism used to verify the hypothesis of Kusakabe's theorem. Identify $\mathbb{C}^3$ with $\mathbb{C} \times \mathbb{C}^2$ by using coordinates (ζ, w_1, w_2) on the target.

Proposition 2.1. *The polynomial map*

$$\Phi: \mathbb{C}^3 \longrightarrow \mathbb{C} \times \mathbb{C}^2, \quad \Phi(z_1, z_2, z_3) = (z_1 + i(z_2^2 + z_3^2), z_2, z_3),$$

is a polynomial automorphism with inverse

$$\Psi(\zeta, w_1, w_2) = (\zeta - i(w_1^2 + w_2^2), w_1, w_2).$$

Moreover, for every $x = (x_1, x_2, x_3) \in \mathbb{R}^3$, if

$$\Phi(x) = (\zeta, w) \in \mathbb{C} \times \mathbb{C}^2,$$

then

$$\|w\| \leq 1 + |\zeta|.$$

Consequently,

$$\Phi(\mathbb{R}^3) \subset \{(\zeta, w) \in \mathbb{C} \times \mathbb{C}^2 : \|w\| \leq 1 + |\zeta|\}.$$

Proof. Direct substitution gives $\Psi \circ \Phi = \text{id}_{\mathbb{C}^3}$ and $\Phi \circ \Psi = \text{id}_{\mathbb{C} \times \mathbb{C}^2}$, so Φ is a polynomial automorphism with inverse Ψ .

Set

$$r = \sqrt{x_2^2 + x_3^2} \geq 0.$$

The definition of Φ gives

$$\zeta = x_1 + i(x_2^2 + x_3^2) = x_1 + ir^2, \quad w = (x_2, x_3).$$

Thus $\|w\| = r$ and $r^2 = \text{Im } \zeta \leq |\zeta|$. Since $r \leq 1 + r^2$ for $r \geq 0$, it follows that

$$\|w\| = r \leq 1 + r^2 \leq 1 + |\zeta|.$$

□

The following is the required form of Kusakabe's theorem for a possibly unbounded closed set [[21](#), Theorem 1.6].

Theorem 2.2. *Suppose that $n > 1$ and that $E \subset \mathbb{C}^n$ is a closed polynomially convex set for which there exist a constant $C > 0$ and a holomorphic automorphism φ of $\mathbb{C}^n$ such that, under the splitting $\mathbb{C}^n = \mathbb{C}^{n-2} \times \mathbb{C}^2$,*

$$\varphi(E) \subset \{(z, w) \in \mathbb{C}^{n-2} \times \mathbb{C}^2 : \|w\| \leq C(1 + \|z\|)\}.$$

Then $\mathbb{C}^n \setminus E$ is Oka.

Proof of Theorem 1.3. Let $S \subset \mathbb{R}^3$ be closed in the relative topology. Since the standard real subspace $\mathbb{R}^3$ is closed in $\mathbb{C}^3$, the set S is closed in $\mathbb{C}^3$.

For each $j \geq 1$, set

$$S_j = S \cap \{z \in \mathbb{C}^3 : \|z\| \leq j\}.$$

Each S_j is a compact subset of $\mathbb{R}^3$ and is therefore polynomially convex by [[9](#), Example 1]. The sequence $\{S_j\}$ is a compact exhaustion of S , so [Definition 1.2](#) gives

$$\widehat{S} = \bigcup_{j=1}^{\infty} \widehat{S}_j = \bigcup_{j=1}^{\infty} S_j = S.$$

Thus S is a closed polynomially convex subset of $\mathbb{C}^3$.

The automorphism and estimate in **Proposition 2.1** give

$$\Phi(S) \subset \Phi(\mathbb{R}^3) \subset \{(\zeta, w) \in \mathbb{C} \times \mathbb{C}^2 : \|w\| \leq 1 + |\zeta|\}.$$

This is the hypothesis of Theorem 2.2 for the splitting $\mathbb{C}^3 = \mathbb{C} \times \mathbb{C}^2$ and $C = 1$. Hence $\mathbb{C}^3 \setminus S$ is Oka. $\square$

3. THE CLOSED HARTOGS TRIANGLE

Set $H = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1| \leq |z_2| \leq 1\}$ and $\Omega = \mathbb{C}^2 \setminus H$. Here $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ denotes the unit disc. The set H is compact. Its complement is connected because

$$\Omega = \{|z_1| > |z_2|\} \cup \{|z_2| > 1\};$$

the first set is biholomorphic to $\mathbb{C}^* \times \mathbb{D}$, the second is $\mathbb{C} \times (\mathbb{C} \setminus \overline{\mathbb{D}})$, and their intersection is nonempty. Therefore Ω is a domain.

For each $a \in \mathbb{D}$, set $L_a = \{(z_1, z_2) \in \mathbb{C}^2 : z_2 = az_1\}$ and $U_a = \Omega \setminus L_a$. The line L_a is closed in $\mathbb{C}^2$, so U_a is open in Ω and hence is a complex manifold. On $\mathbb{C}^2 \setminus L_a$, consider the map

$$\Psi_a : \mathbb{C}^2 \setminus L_a \longrightarrow \mathbb{C} \times \mathbb{C}^*, \quad \Psi_a(z_1, z_2) = \left(\frac{z_1}{z_2 - az_1}, z_2 - az_1 \right) = (u, v).$$

Here $v = z_2 - az_1$ vanishes precisely on L_a . Under the simultaneous scaling $(z_1, z_2) \mapsto (\lambda z_1, \lambda z_2)$ with $\lambda \in \mathbb{C}^*$, both z_1 and v are multiplied by λ , so $u = z_1/v$ is unchanged and therefore depends only on the projective direction $[z_1 : z_2]$. On the chart $\mathbb{P}^1 \setminus \{[1 : a]\}$, consisting exactly of the directions not represented by L_a , it is an affine coordinate with inverse $u \mapsto [u : 1 + au]$. Thus Ψ_a records the projective direction in the coordinate u and the nonzero scalar in the coordinate v . The next lemma determines its image on U_a .

Lemma 3.1. *For every $a \in \mathbb{D}$, the set*

$$K_a = \{(u, v) \in \mathbb{C}^2 : |u| \leq |1 + au|, |(1 + au)v| \leq 1\}$$

is compact and polynomially convex, and Ψ_a restricts to a biholomorphism from U_a onto $(\mathbb{C} \times \mathbb{C}^) \setminus K_a$.*

Proof. Fix $a \in \mathbb{D}$. First determine the image of $H \setminus L_a$ under Ψ_a . On $\mathbb{C}^2 \setminus L_a$ one has $z_2 - az_1 \neq 0$, so the functions

$$u = \frac{z_1}{z_2 - az_1}, \quad v = z_2 - az_1$$

are well defined and holomorphic. They satisfy

$$z_1 = uv, \quad z_2 = v + az_1 = (1 + au)v.$$

Set

$$\Theta_a : \mathbb{C} \times \mathbb{C}^* \longrightarrow \mathbb{C}^2 \setminus L_a, \quad \Theta_a(u, v) = (uv, (1 + au)v).$$

For every $(u, v) \in \mathbb{C} \times \mathbb{C}^*$, its image satisfies

$$(1 + au)v - a(uv) = v \neq 0,$$

so $\Theta_a(u, v) \notin L_a$. Direct substitution into the formulas for Ψ_a and Θ_a gives

$$\Psi_a \circ \Theta_a = \text{id}_{\mathbb{C} \times \mathbb{C}^*}, \quad \Theta_a \circ \Psi_a = \text{id}_{\mathbb{C}^2 \setminus L_a}.$$

Thus Ψ_a is a biholomorphism.

Let $(u, v) \in \mathbb{C} \times \mathbb{C}^*$, and set $(z_1, z_2) = \Theta_a(u, v)$. Since $v \neq 0$, the two defining inequalities for membership in H become

$$|z_1| \leq |z_2| \iff |uv| \leq |(1 + au)v| \iff |u| \leq |1 + au|, \quad |z_2| \leq 1 \iff |(1 + au)v| \leq 1.$$

Moreover, $L_a \cap H = \{0\}$, so $H \setminus L_a = H \setminus \{0\}$. The equivalences above therefore show that

$$\Psi_a(H \setminus \{0\}) = K_a \cap (\mathbb{C} \times \mathbb{C}^*). \quad (3.1)$$

Combining the definition of U_a with (3.1) gives

$$\begin{aligned} \Psi_a(U_a) &= \Psi_a((\mathbb{C}^2 \setminus L_a) \setminus (H \setminus \{0\})) \\ &= (\mathbb{C} \times \mathbb{C}^*) \setminus (K_a \cap (\mathbb{C} \times \mathbb{C}^*)) \\ &= (\mathbb{C} \times \mathbb{C}^*) \setminus K_a. \end{aligned} \quad (3.2)$$

The last equality is an equality of subsets of $\mathbb{C} \times \mathbb{C}^*$: points of K_a with $v = 0$ do not belong to that ambient manifold and hence do not affect the set difference.

It remains to prove that K_a is compact and polynomially convex. The identity

$$|(1 - |a|^2)u - \bar{a}|^2 - 1 = (1 - |a|^2)(|u|^2 - |1 + au|^2).$$

Since $1 - |a|^2 > 0$, this gives the equivalence

$$|u| \leq |1 + au| \iff |(1 - |a|^2)u - \bar{a}| \leq 1.$$

Consequently,

$$K_a = \{(u, v) \in \mathbb{C}^2 : |(1 - |a|^2)u - \bar{a}| \leq 1, |(1 + au)v| \leq 1\}. \quad (3.3)$$

For $(u, v) \in K_a$, the triangle inequality gives directly

$$|u| \leq |1 + au| \leq 1 + |a||u|, \quad \text{hence} \quad |u| \leq \frac{1}{1 - |a|}. \quad (3.4)$$

Moreover, the inequality $|u| \leq |1 + au|$ gives

$$1 = |(1 + au) - au| \leq |1 + au| + |a||u| \leq (1 + |a|)|1 + au|.$$

It follows that $|1 + au| \geq 1/(1 + |a|)$, and the inequality $|(1 + au)v| \leq 1$ gives $|v| \leq 1 + |a|$. Together with (3.4), this shows that K_a is bounded and, being closed, compact.

For the fixed parameter a , set

$$p_a(u, v) = (1 - |a|^2)u - \bar{a}, \quad q_a(u, v) = (1 + au)v.$$

Then $p_a, q_a \in \mathbb{C}[u, v]$, and (3.3) becomes

$$K_a = \{(u, v) \in \mathbb{C}^2 : |p_a(u, v)| \leq 1, |q_a(u, v)| \leq 1\}.$$

Accordingly, K_a is a compact polynomial polyhedron and is therefore polynomially convex by [19, Lemma 2.7.4]. Together with (3.2), this proves the lemma. $\square$

Corollary 3.2. *For every $a \in \mathbb{D}$, the complex manifold U_a is Oka.*

Proof. By Lemma 3.1, the manifold U_a is biholomorphic to $(\mathbb{C} \times \mathbb{C}^*) \setminus K_a$ for a compact polynomially convex set $K_a \subset \mathbb{C}^2$. Forstnerič and Fornæss Wold proved that if K is a compact polynomially convex set in $\mathbb{C}^N$ for some $N > 1$, then $(\mathbb{C}^{N-1} \times \mathbb{C}^*) \setminus K$ is an Oka manifold [17, Corollary 3.2]. Applying this theorem with $N = 2$ and $K = K_a$ therefore shows that U_a is Oka. $\square$

Proof of Theorem 1.4. By Corollary 3.2, the two subsets

$$U_0 = \Omega \setminus L_0, \quad U_{1/2} = \Omega \setminus L_{1/2}$$

are Zariski-open Oka manifolds, and $\Omega = U_0 \cup U_{1/2}$. Then every point of Ω has a Zariski-open Oka neighborhood. Kusakabe's localization theorem [20, Theorem 1.4] states that if Y is a connected complex manifold such that every point of Y has a Zariski-open Oka neighborhood, then Y is an Oka manifold. Applying this theorem with $Y = \Omega$ therefore shows that Ω is Oka, which proves Theorem 1.4. $\square$

4. BLOW-UPS ALONG CONNECTED OKA CENTERS

The proof of Theorem 1.6 rests on a closed discrete set $D \subset \mathbb{C}^n$ such that, for every nondegenerate entire map $F: \mathbb{C}^n \rightarrow \mathbb{C}^n$, there is a point $z \in \mathbb{C}^n$ with $F(z) \in D$ and $\text{Jac } F(z) \neq 0$. Rosay and Rudin obtained unavoidable discrete sets by choosing finite subsets of expanding spheres [25, Lemmas 4.3–4.4 and Theorem 4.5]. The argument below chooses these finite sets so that the required intersection occurs at a regular point.

Theorem 4.1. *For every integer $n > 1$, there is a nonempty closed discrete set $D \subset \mathbb{C}^n$ such that*

$$F^{-1}(D) \not\subset \text{Crit}(F)$$

for every nondegenerate entire map $F: \mathbb{C}^n \rightarrow \mathbb{C}^n$.

Proof of Theorem 4.1. Fix $n > 1$, and write $\mathbb{B}^n$ for the unit ball in $\mathbb{C}^n$. Choose numbers $0 < a_1 < a_2$, $0 < r_1 < r_2$, and $c > 0$. Let $\mathcal{F}$ be the family of holomorphic maps $G: a_2\mathbb{B}^n \rightarrow r_2\mathbb{B}^n$ such that

$$\|G(0)\| < \frac{r_1}{2}, \quad |\text{Jac } G| \geq c \quad \text{at some point of } a_1\mathbb{B}^n.$$

Choose nonempty finite sets $E_1 \subset E_2 \subset \cdots \subset \partial(r_1\mathbb{B}^n)$ whose union is dense in $\partial(r_1\mathbb{B}^n)$. We claim that, for some ℓ , every $G \in \mathcal{F}$ satisfies

$$G(a_1\mathbb{B}^n) \cap \partial(r_1\mathbb{B}^n) \neq \emptyset \implies G^{-1}(E_\ell) \not\subset \text{Crit}(G).$$

If not, for every ℓ there are $F_\ell \in \mathcal{F}$ and $z_\ell \in a_1\mathbb{B}^n$ such that

$$F_\ell(z_\ell) \in \partial(r_1\mathbb{B}^n), \quad F_\ell^{-1}(E_\ell) \subset \text{Crit}(F_\ell).$$

Choose $\xi_\ell \in a_1\mathbb{B}^n$ with $|\text{Jac } F_\ell(\xi_\ell)| \geq c$. The sets $\overline{a_1\mathbb{B}^n}$ and $\overline{r_2\mathbb{B}^n}$ are compact, and the maps F_ℓ are uniformly bounded by r_2 . By successively passing to subsequences, compactness and Montel's theorem give three subsequences, selected along the same original indices; the corresponding finite sets E_ℓ are retained as well. Relabeling these subsequences and finite sets by ℓ , we have

$$z_\ell \longrightarrow w \in \overline{a_1\mathbb{B}^n}, \quad \xi_\ell \longrightarrow \xi \in \overline{a_1\mathbb{B}^n}, \quad F_\ell \longrightarrow F \quad \text{uniformly on compact subsets of } a_2\mathbb{B}^n,$$

where $F: a_2\mathbb{B}^n \rightarrow \mathbb{C}^n$ is holomorphic and $F(a_2\mathbb{B}^n) \subset \overline{r_2\mathbb{B}^n}$. Since $\overline{a_1\mathbb{B}^n} \subset a_2\mathbb{B}^n$, the convergence is uniform on $\overline{a_1\mathbb{B}^n}$, and therefore

$$\|F_\ell(z_\ell) - F(w)\| \leq \|F_\ell(z_\ell) - F(z_\ell)\| + \|F(z_\ell) - F(w)\| \longrightarrow 0.$$

Cauchy estimates give uniform convergence of the first derivatives, and hence of the Jacobian determinants, on $\overline{a_1\mathbb{B}^n}$. Thus

$$|\text{Jac } F_\ell(\xi_\ell) - \text{Jac } F(\xi)| \leq |\text{Jac } F_\ell(\xi_\ell) - \text{Jac } F(\xi_\ell)| + |\text{Jac } F(\xi_\ell) - \text{Jac } F(\xi)| \longrightarrow 0.$$

It follows that

$$F(w) \in \partial(r_1\mathbb{B}^n), \quad |\text{Jac } F(\xi)| \geq c.$$

In particular, F is nondegenerate and $\|F(0)\| \leq r_1/2$.

Set $Q = \{z \in a_2\mathbb{B}^n : \text{Jac } F(z) \neq 0\}$. Since $\text{Jac } F \not\equiv 0$, its zero set is a proper analytic subset of $a_2\mathbb{B}^n$. Its complement Q is connected and dense by [6, Chapter II, Remark 4.2]. Density and $\|F(0)\| \leq r_1/2$ give $F(Q) \cap r_1\mathbb{B}^n \neq \emptyset$. If $F(Q) \subset \overline{r_1\mathbb{B}^n}$, density gives $F(a_2\mathbb{B}^n) \subset \overline{r_1\mathbb{B}^n}$. Let $e = F(w)/r_1$ and $g(z) = \langle F(z), e \rangle$, where the standard Hermitian inner product is taken to be linear in the first variable. Since $\|e\| = 1$, the Cauchy–Schwarz inequality gives

$$|g(z)| \leq \|F(z)\| \leq r_1, \quad g(w) = r_1.$$

The maximum modulus principle gives $g \equiv r_1$. Hence, for every $z \in a_2\mathbb{B}^n$,

$$r_1 = |\langle F(z), e \rangle| \leq \|F(z)\| \leq r_1.$$

Equality in Cauchy–Schwarz and $g(z) = r_1$ now give $F(z) = r_1 e = F(w)$. Thus F is constant, contradicting $|\text{Jac } F(\xi)| \geq c$. Therefore $F(Q)$ also meets $\mathbb{C}^n \setminus \overline{r_1\mathbb{B}^n}$, and its connectedness gives

$$F(Q) \cap \partial(r_1\mathbb{B}^n) \neq \emptyset.$$

Choose $p \in Q$ and $q = F(p) \in \partial(r_1\mathbb{B}^n)$. There are neighborhoods U of p and V of q such that $F: U \rightarrow V$ is biholomorphic. Choose $\delta > 0$ with $\overline{B(q, \delta)} \subset V$, write $\phi = (F|_U)^{-1}: V \rightarrow U$, and consider the holomorphic maps

$$h_\ell = F_\ell \circ \phi: V \longrightarrow \mathbb{C}^n.$$

The set $K = \phi(\overline{B(q, \delta)})$ is a compact subset of U . The convergence $F_\ell \rightarrow F$ and the Cauchy estimates give uniform convergence of both F_ℓ and dF_ℓ on K . Since $d\phi$ is bounded on $\overline{B(q, \delta)}$, the identities

$$h_\ell(y) - y = (F_\ell - F)(\phi(y)), \quad dh_\ell(y) - I = (dF_\ell - dF)_{\phi(y)} \circ d\phi_y$$

show that $h_\ell \rightarrow \text{id}$ in the $\mathcal{C}^1$ topology on $\overline{B(q, \delta)}$. For all sufficiently large ℓ ,

$$\sup_{\overline{B(q, \delta)}} \|h_\ell - \text{id}\| < \frac{\delta}{4}, \quad \sup_{\overline{B(q, \delta)}} \|dh_\ell - I\| < \frac{1}{2}.$$

Here $\|h_\ell(y) - y\|$ is the Euclidean norm on $\mathbb{C}^n$, $\|dh_\ell(y) - I\|$ is the induced operator norm, and I is the identity map on $\mathbb{C}^n$. If $dh_\ell(y)v = 0$, then

$$\|v\| = \|(I - dh_\ell(y))v\| \leq \|I - dh_\ell(y)\| \|v\| < \frac{1}{2} \|v\|,$$

so $v = 0$. Thus $dh_\ell(y)$ is injective and hence, as an endomorphism of the finite-dimensional vector space $\mathbb{C}^n$, invertible throughout $\overline{B(q, \delta)}$. For $y_0 \in \overline{B(q, \delta/2)}$, the map $T_{\ell, y_0}(y) = y + y_0 - h_\ell(y)$ satisfies

$$\|T_{\ell, y_0}(y) - q\| < \frac{3\delta}{4}, \quad \|dT_{\ell, y_0}(y)\| < \frac{1}{2} \quad (y \in \overline{B(q, \delta)}).$$

The first estimate gives $T_{\ell, y_0}(\overline{B(q, \delta)}) \subset B(q, 3\delta/4) \subset \overline{B(q, \delta)}$, while the second and the convexity of the ball give $\|T_{\ell, y_0}(y) - T_{\ell, y_0}(y')\| \leq \frac{1}{2}\|y - y'\|$. The Banach fixed-point theorem therefore gives $y \in \overline{B(q, \delta)}$ with $T_{\ell, y_0}(y) = y$, or equivalently $h_\ell(y) = y_0$.

The reindexed sets E_ℓ are still increasing and have dense union in $\partial(r_1 \mathbb{B}^n)$. For every sufficiently large ℓ , choose $e_\ell \in E_\ell \cap B(q, \delta/2)$. Taking $y_0 = e_\ell$ in the fixed-point conclusion gives $y_\ell \in \overline{B(q, \delta)}$ with $h_\ell(y_\ell) = e_\ell$. Set $x_\ell = \phi(y_\ell)$. Then $F_\ell(x_\ell) = e_\ell$, while $dh_\ell(y_\ell) = dF_\ell(x_\ell) \circ d\phi(y_\ell)$ and the invertibility of $dh_\ell(y_\ell)$ and $d\phi(y_\ell)$ give $\text{Jac } F_\ell(x_\ell) \neq 0$. This contradicts $F_\ell^{-1}(E_\ell) \subset \text{Crit}(F_\ell)$ and proves the claim.

For each positive integer t , we next construct a discrete set $E_t \subset \mathbb{C}^n \setminus t\mathbb{B}^n$ with a scale- t growth property. Fix t and take sequences

$$\frac{t}{2} = a_1 < a_2 < \cdots \longrightarrow \frac{3t}{4}, \quad t = r_1 < r_2 < \cdots \longrightarrow \infty.$$

For every j , choose nonempty finite sets

$$E_{t,j,1} \subset E_{t,j,2} \subset \cdots \subset \partial(r_j \mathbb{B}^n)$$

whose union is dense in $\partial(r_j \mathbb{B}^n)$. Apply the claim with

$$(a_1, a_2, r_1, r_2, c) = (a_j, a_{j+1}, r_j, r_{j+1}, 1/t)$$

and with the auxiliary sequence $E_\ell = E_{t,j,\ell}$. There is an index $\ell_{t,j}$ for which the conclusion of the claim holds. Write $E_{t,j} = E_{t,j,\ell_{t,j}}$. Consequently, every holomorphic map $\Phi: a_{j+1}\mathbb{B}^n \rightarrow r_{j+1}\mathbb{B}^n$ satisfying

$$\|\Phi(0)\| < \frac{r_j}{2}, \quad |\text{Jac } \Phi(z)| \geq \frac{1}{t} \quad \text{for some } z \in a_j \mathbb{B}^n$$

also satisfies

$$\Phi(a_j \mathbb{B}^n) \cap \partial(r_j \mathbb{B}^n) \neq \emptyset \implies \Phi^{-1}(E_{t,j}) \not\subset \text{Crit}(\Phi).$$

Set

$$E_t = \bigcup_{j=1}^{\infty} E_{t,j}.$$

Since $E_{t,j} \subset \partial(r_j \mathbb{B}^n)$ and $r_j \rightarrow \infty$, the set E_t is nonempty, closed, and discrete, and $E_t \subset \mathbb{C}^n \setminus t\mathbb{B}^n$.

Let $\Psi: t\mathbb{B}^n \rightarrow \mathbb{C}^n$ be holomorphic and suppose that

$$\|\Psi(0)\| < \frac{t}{2}, \quad |\text{Jac } \Psi(\zeta)| \geq \frac{1}{t} \quad \text{for some } \zeta \in \frac{t}{2}\mathbb{B}^n, \quad \Psi^{-1}(E_t) \subset \text{Crit}(\Psi).$$

For example, one may take Ψ to be the identity map. We claim that

$$\Psi\left(\frac{t}{2}\mathbb{B}^n\right) \subset t\mathbb{B}^n.$$

The proof of [25, Lemma 4.4] shows that the growth estimate follows by downward induction once one has a starting inclusion and can pass from the inclusion at level $k+1$ to the one at level k . Choose J such that

$$r_{J+1} > M := \max_{z \in (3t/4)\mathbb{B}^n} \|\Psi(z)\|.$$

Since $a_{J+1} < 3t/4$, this gives

$$\Psi(a_{J+1}\mathbb{B}^n) \subset r_{J+1}\mathbb{B}^n.$$

Fix $1 \leq k \leq J$ and suppose that $\Psi(a_{k+1}\mathbb{B}^n) \subset r_{k+1}\mathbb{B}^n$. The restriction $\Psi|_{a_{k+1}\mathbb{B}^n}: a_{k+1}\mathbb{B}^n \rightarrow r_{k+1}\mathbb{B}^n$ is holomorphic and, since $r_k \geq t$ and $a_k \geq t/2$, satisfies

$$\|\Psi(0)\| < \frac{r_k}{2}, \quad |\text{Jac } \Psi(\zeta)| \geq \frac{1}{t}, \quad \zeta \in a_k \mathbb{B}^n.$$

Our choice of $E_{t,k}$ therefore gives the stronger implication

$$\Psi(a_k \mathbb{B}^n) \cap \partial(r_k \mathbb{B}^n) \neq \emptyset \implies \Psi^{-1}(E_{t,k}) \not\subset \text{Crit}(\Psi).$$

The inclusion $E_{t,k} \subset E_t$ and the condition $\Psi^{-1}(E_t) \subset \text{Crit}(\Psi)$ rule out $\Psi^{-1}(E_{t,k}) \not\subset \text{Crit}(\Psi)$. Hence $\Psi(a_k \mathbb{B}^n)$ avoids $\partial(r_k \mathbb{B}^n)$. Since this image is connected and contains $\Psi(0) \in r_k \mathbb{B}^n$, it follows that

$$\Psi(a_{k+1} \mathbb{B}^n) \subset r_{k+1} \mathbb{B}^n \implies \Psi(a_k \mathbb{B}^n) \subset r_k \mathbb{B}^n.$$

Starting with the inclusion at level $J+1$ and applying this implication for $k = J, J-1, \dots, 1$ gives $\Psi((t/2) \mathbb{B}^n) \subset t \mathbb{B}^n$, as claimed.

Set

$$D = \bigcup_{t=1}^{\infty} E_t.$$

For every $R > 0$, the sets E_t with $t \geq R$ do not meet $R \mathbb{B}^n$, while each of the finitely many sets E_t with $t < R$ meets $R \mathbb{B}^n$ in a finite set. It follows that $D \cap R \mathbb{B}^n$ is finite, so D is nonempty, closed, and discrete.

Let $F: \mathbb{C}^n \rightarrow \mathbb{C}^n$ be nondegenerate. If $F^{-1}(D) \subset \text{Crit}(F)$, take $z_0 \in \mathbb{C}^n$ such that $\text{Jac } F(z_0) \neq 0$ and write $\eta = |\text{Jac } F(z_0)| > 0$. Since $E_t \subset D$, for every sufficiently large integer t ,

$$\|F(0)\| < \frac{t}{2}, \quad z_0 \in \frac{t}{2} \mathbb{B}^n, \quad \eta \geq \frac{1}{t}, \quad F^{-1}(E_t) \subset \text{Crit}(F).$$

The scale- t growth estimate with $\Psi = F|_{t \mathbb{B}^n}$ gives

$$F\left(\frac{t}{2} \mathbb{B}^n\right) \subset t \mathbb{B}^n \quad (t \gg 1).$$

The proof of [25, Theorem 4.5] shows that the inclusions $F((t/2) \mathbb{B}^n) \subset t \mathbb{B}^n$ for all sufficiently large positive integers t imply that F is affine. Write $F(z) = Az + b$. Nondegeneracy gives $\det A \neq 0$, so F is surjective and $\text{Crit}(F) = \emptyset$. Take $d \in D$. Then

$$F^{-1}(d) \neq \emptyset, \quad F^{-1}(d) \cap \text{Crit}(F) = \emptyset,$$

contrary to $F^{-1}(D) \subset \text{Crit}(F)$. □

Proof of Theorem 1.6. Fix $n \geq 3$, and choose $D \subset \mathbb{C}^n$ as in Theorem 4.1. The construction shows that D meets every compact subset of $\mathbb{C}^n$ in a finite set and is unbounded. Its distinct points can therefore be enumerated as a sequence $\{a_j\}$ with $\|a_j\| \rightarrow \infty$. By [16, Proposition 2], there is a proper holomorphic embedding

$$\gamma: \mathbb{C} \hookrightarrow \mathbb{C}^n$$

whose image $A = \gamma(\mathbb{C})$ contains the sequence $\{a_j\}$, hence contains D . Properness makes A closed; moreover, A is connected, smooth, and biholomorphic to $\mathbb{C}$. Its codimension is $n-1 \geq 2$.

Write

$$\pi: Y = \text{Bl}_A \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

for the blow-down map. If there is a nondegenerate holomorphic map $f: \mathbb{C}^n \rightarrow Y$, choose $z_0 \in \mathbb{C}^n$ where df_{z_0} is invertible. By the holomorphic inverse function theorem, there is a neighborhood U of z_0 such that $f|_U: U \rightarrow f(U)$ is biholomorphic and $f(U)$ is open in Y .

Set $E = \pi^{-1}(A)$. By [6, Chapter VII, p. 354], the set E is a smooth hypersurface and $\pi: Y \setminus E \rightarrow \mathbb{C}^n \setminus A$ is biholomorphic; in particular, E has empty interior. Fix $y \in E$ and write $s = \text{codim}_{\mathbb{C}^n} A = n-1$. Choose local coordinates $(z_1, \dots, z_n)$ near $\pi(y)$ such that $A = \{z_1 = \dots = z_s = 0\}$, and a blow-up chart containing y , indexed by j , in which $E = \{w_j = 0\}$. In these coordinates, [6, Chapter VII, formula (12.2)] gives

$$\pi(w_1, \dots, w_n) = (w_1 w_j, \dots, w_{j-1} w_j, w_j, w_{j+1} w_j, \dots, w_s w_j, w_{s+1}, \dots, w_n).$$

At y , where $w_j = 0$, this formula annihilates $\partial/\partial w_k$ for $1 \leq k \leq s$, $k \neq j$. The image of $\partial/\partial w_j$ has $\partial/\partial z_j$ -component 1, while $d\pi_y(\partial/\partial w_\ell) = \partial/\partial z_\ell$ for $\ell > s$. These $n-s+1$ images are linearly independent, and hence

$$\text{rank } d\pi_y = n-s+1 = \dim A + 1 = 2.$$

Since $f(U)$ is a nonempty open subset of Y and E has empty interior, $f(U) \setminus E \neq \emptyset$. Choose $y_1 \in f(U) \setminus E$ and let $\zeta_1 = (f|_U)^{-1}(y_1)$. The differentials df_{ζ_1} and $d\pi_{y_1}$ are invertible, so

$$F = \pi \circ f: \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

is nondegenerate. On the other hand, if $z \in F^{-1}(A)$, then $f(z) \in E$, and the displayed rank formula gives

$$\text{rank } dF_z \leq \text{rank } d\pi_{f(z)} = 2 < n.$$

Thus $F^{-1}(A) \subset \text{Crit}(F)$ and, since $D \subset A$, $F^{-1}(D) \subset \text{Crit}(F)$. This contradicts [Theorem 4.1](#). Therefore Y is Brody volume hyperbolic and, by [\[15, p. 770\]](#), is not Oka. $\square$

5. THE ALGEBRAIC BASIC OKA PROBLEM

It remains to prove [Theorem 1.8](#). All cohomology groups in this section are singular cohomology groups with rational coefficients unless otherwise specified. We first construct a continuous map with nonzero pullback and then compare Hodge types and weights to show that no regular map can have the same pullback.

Consider the projective cubic $E = \{[X : Y : Z] \in \mathbb{P}^2 : Y^2Z = X^3 - XZ^2\}$, denote its point at infinity by $p = [0 : 1 : 0]$, and set $E^\circ = E \setminus \{p\}$. On $Z = 0$ the equation gives $X = 0$, so p is the unique point of E at infinity. For

$$\Phi(X, Y, Z) = Y^2Z - X^3 + XZ^2$$

one has

$$\Phi_X = -3X^2 + Z^2, \quad \Phi_Y = 2YZ, \quad \Phi_Z = Y^2 + 2XZ.$$

On $Z = 1$, a singular point would satisfy $y = 0$ and $x^2 = 1/3$, which is incompatible with $y^2 = x^3 - x$. Thus the affine curve is nonsingular, while $\Phi_Z(p) = 1$ shows that the unique point p of E at infinity is also nonsingular. Hence E is nonsingular. Moreover, $x^3 - x$ has the simple zeros $0, 1, -1$ and is therefore not a square in $\mathbb{C}(x)$. Thus $y^2 - (x^3 - x)$ is irreducible in $\mathbb{C}(x)[y]$, and hence in $\mathbb{C}[x, y]$ by Gauss's lemma; its homogenization Φ is also irreducible. Consequently, E is irreducible, and

$$E^\circ = \{(x, y) \in \mathbb{C}^2 : y^2 = x^3 - x\}.$$

Fischer's genus formula [\[8, p. 177\]](#) gives $g(E) = 1$. Hence E is homeomorphic to a torus. Lawson's construction [\[24, p. 209\]](#) provides a deformation retraction of E° onto an embedded wedge $S^1 \vee S^1$; denote its terminal retraction by $h: E^\circ \rightarrow S^1 \vee S^1$.

Fix an integer $n \geq 2$, and set $X_n = E^\circ \times (\mathbb{C}^*)^{2n-2}$ and $Y_n = \mathbb{C}^n \setminus \{0\}$. The curve E° is affine by its displayed equation, and $\mathbb{C}^* = \text{Spec } \mathbb{C}[t, t^{-1}]$ is affine. The Jacobian calculation above shows that E° is smooth, and each factor $\mathbb{C}^*$ is smooth. Their product X_n is therefore a smooth affine algebraic variety of complex dimension $1 + (2n - 2) = 2n - 1$. The deformation retraction h shows that E° is path connected, and every $\mathbb{C}^*$ -factor is path connected as well. Hence X_n is a smooth connected affine algebraic variety and is an admissible source under both aBOP conventions discussed in the Introduction. The target Y_n is a smooth connected algebraic variety, being the complement of one point in $\mathbb{A}_{\mathbb{C}}^n$ with $n \geq 2$.

Lemma 5.1. *For the fixed integer $n \geq 2$, there is a continuous map $F_n: X_n \rightarrow Y_n$ whose pullback in degree $2n - 1$ is nonzero.*

Proof. Let $\lambda: S^1 \vee S^1 \rightarrow S^1$ be the map that is the identity on the first circle and collapses the second circle to the wedge point. Denote by $s: S^1 \rightarrow E^\circ$ the inclusion of the first circle and set $\rho = \lambda \circ h$. Give S^1 its standard orientation, and let $\alpha \in H^1(S^1; \mathbb{Q})$ be the rational cohomology orientation class normalized by $\langle \alpha, [S^1] \rangle = 1$. The identity $\rho \circ s = \text{id}_{S^1}$ gives

$$s^*(\rho^* \alpha) = (\rho \circ s)^* \alpha = \alpha \neq 0,$$

so $\rho^* \alpha \neq 0$.

For each $\mathbb{C}^*$ -factor, the maps $R_t: \mathbb{C}^* \rightarrow \mathbb{C}^*$, $R_t(z) = ((1 - t) + t/|z|)z$, form a deformation retraction onto S^1 . Its terminal retraction is

$$r: \mathbb{C}^* \rightarrow S^1, \quad r(z) = \frac{z}{|z|}$$

at $t = 1$. Thus $r^*: H^1(S^1; \mathbb{Q}) \rightarrow H^1(\mathbb{C}^*; \mathbb{Q})$ is an isomorphism. The maps ρ and r give

$$q_n: X_n \rightarrow T^{2n-1} = (S^1)^{2n-1}, \quad q_n(x, z_1, \dots, z_{2n-2}) = (\rho(x), r(z_1), \dots, r(z_{2n-2})).$$

To construct a degree-one map from T^{2n-1} to S^{2n-1} , give the sphere its standard orientation and the torus its product orientation. Let $u_n \in H^{2n-1}(S^{2n-1}; \mathbb{Z})$ and $\theta_n \in H^{2n-1}(T^{2n-1}; \mathbb{Z})$ be the resulting cohomology orientation classes. Write $\pi_0: X_n \rightarrow E^\circ$ and $\pi_j: X_n \rightarrow \mathbb{C}^*$, $1 \leq j \leq 2n - 2$, for the factor projections. Viewing θ_n with rational coefficients, the product orientation, the Künneth theorem [\[18, Theorem 3.15\]](#), and the nonvanishing of $\rho^* \alpha$ and $r^* \alpha$ give

$$q_n^* \theta_n = \pi_0^*(\rho^* \alpha) \smile \pi_1^*(r^* \alpha) \smile \dots \smile \pi_{2n-2}^*(r^* \alpha) \neq 0.$$

The standard product CW structure makes T^{2n-1} a $(2n-1)$ -dimensional CW complex. Since $2n-1 \geq 3$, the Hopf classification theorem [1, Chapter V, Theorems 11.6 and 11.9] gives a continuous map

$$c_n: T^{2n-1} \longrightarrow S^{2n-1}$$

such that $c_n^* u_n = \theta_n$; equivalently, c_n has degree 1. Identify S^{2n-1} with the unit sphere in $\mathbb{C}^n$, and denote its inclusion into Y_n by ι_n . The radial deformation retraction $z \mapsto z/\|z\|$ makes ι_n a homotopy equivalence. After extending u_n to rational coefficients, choose $\eta_n \in H^{2n-1}(Y_n; \mathbb{Q})$ with $\iota_n^* \eta_n = u_n$. The composite

$$F_n = \iota_n \circ c_n \circ q_n: X_n \longrightarrow Y_n$$

satisfies

$$F_n^* \eta_n = q_n^* c_n^* \iota_n^* \eta_n = q_n^* \theta_n \neq 0.$$

Thus $F_n^*: H^{2n-1}(Y_n; \mathbb{Q}) \rightarrow H^{2n-1}(X_n; \mathbb{Q})$ is nonzero. $\square$

To show that no regular map is homotopic to F_n , we compare the Hodge types and weights of these two cohomology groups. For every smooth complex algebraic variety Z , Deligne's canonical splitting has the form

$$H^k(Z; \mathbb{C}) = \bigoplus_{p,q} I^{p,q} H^k(Z).$$

The symbol $I^{p,q} H^k(Z)$ denotes the component of Hodge bidegree (p, q) ; such a component has Hodge type (p, q) and weight $p+q$. The weight filtration on the cohomology group is the increasing filtration given by

$$W_\ell H^k(Z; \mathbb{C}) = \bigoplus_{p+q \leq \ell} I^{p,q} H^k(Z), \quad W_\ell H^k(Z; \mathbb{Q}) = H^k(Z; \mathbb{Q}) \cap W_\ell H^k(Z; \mathbb{C})$$

[4, Lemma 1.2.11]. Thus $W_\ell H^k(Z; \mathbb{Q})$ consists precisely of the rational cohomology classes whose complex Hodge components all have weight at most ℓ . For smooth Z , the possible weights in $H^k(Z; \mathbb{Q})$ range from k to $2k$ [4, Corollary 3.2.15], so

$$0 = W_{k-1} H^k(Z; \mathbb{Q}) \subseteq W_k H^k(Z; \mathbb{Q}) \subseteq \cdots \subseteq W_{2k} H^k(Z; \mathbb{Q}) = H^k(Z; \mathbb{Q}).$$

The group $H^k(Z; \mathbb{Q})$ is pure of weight w when $I^{p,q} H^k(Z) = 0$ for every $p+q \neq w$; equivalently, $W_{w-1} H^k(Z; \mathbb{Q}) = 0$ and $W_w H^k(Z; \mathbb{Q}) = H^k(Z; \mathbb{Q})$. In this pure case one writes $H^{p,q}(Z) = I^{p,q} H^k(Z)$, and

$$H^k(Z; \mathbb{C}) = \bigoplus_{p+q=w} H^{p,q}(Z)$$

is the usual Hodge decomposition; thus $H^{p,q}(Z)$ denotes its component of Hodge type (p, q) . In particular, this is the standard notation for the Hodge decomposition of a smooth projective variety.

Following the rational Tate convention associated with [4, Definition 2.1.13], $\mathbb{Q}(-1)$ denotes the one-dimensional rational Hodge structure whose complexification has Hodge type $(1, 1)$ and weight 2. For a rational Hodge structure V , its Tate twist is $V(-1) = V \otimes_{\mathbb{Q}} \mathbb{Q}(-1)$; it shifts every Hodge type (p, q) to $(p+1, q+1)$ and increases the weight by 2. For $m \geq 1$, write $\mathbb{Q}(-m) = \mathbb{Q}(-1)^{\otimes m}$; twisting by $(-m)$ shifts each Hodge type (p, q) to $(p+m, q+m)$ and increases the weight by $2m$.

Proposition 5.2. *For the fixed integer $n \geq 2$, the group $H^{2n-1}(Y_n; \mathbb{Q})$ is one-dimensional of Hodge type (n, n) and is pure of weight $2n$. Moreover, $H^{2n-1}(X_n; \mathbb{C})$ is the direct sum of two one-dimensional Hodge components of types $(2n-1, 2n-2)$ and $(2n-2, 2n-1)$, and $H^{2n-1}(X_n; \mathbb{Q})$ is pure of weight $4n-3$.*

Proof. The radial deformation retraction of Y_n onto S^{2n-1} gives

$$H^j(Y_n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z}, & j = 0, 2n-1, \\ 0, & \text{otherwise.} \end{cases}$$

Since these integral cohomology groups are free, the universal coefficient theorem [18, Theorem 3.2] gives

$$H^j(Y_n; \mathbb{Q}) \cong H^j(Y_n; \mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Q} \cong \begin{cases} \mathbb{Q}, & j = 0, 2n-1, \\ 0, & \text{otherwise.} \end{cases}$$

The Gysin exact sequence for the smooth closed subvariety $\{0\} \subset \mathbb{A}_{\mathbb{C}}^n$ of codimension n [2, Proposition 2.134] contains the exact sequence of mixed Hodge structures

$$0 = H^{2n-1}(\mathbb{A}_{\mathbb{C}}^n; \mathbb{Q}) \longrightarrow H^{2n-1}(Y_n; \mathbb{Q}) \longrightarrow H^0(\{0\}; \mathbb{Q})(-n) \longrightarrow H^{2n}(\mathbb{A}_{\mathbb{C}}^n; \mathbb{Q}) = 0.$$

Consequently, there is an isomorphism of mixed Hodge structures $H^{2n-1}(Y_n; \mathbb{Q}) \cong \mathbb{Q}(-n)$. Thus the entire group $H^{2n-1}(Y_n; \mathbb{Q})$ has Hodge type (n, n) and is pure of weight $2n$.

Next compute the Hodge types and weights contributed by the factors of X_n . The deformation retraction h cited above gives $H^j(E^\circ; \mathbb{Q}) = 0$ for $j > 1$. Applied to $E^\circ = E \setminus \{p\}$, [2, Example 2.136] gives an isomorphism of mixed Hodge structures $H^1(E^\circ; \mathbb{Q}) \cong H^1(E; \mathbb{Q})$. By [3, Example 1.5.17], $\dim_{\mathbb{C}} H^{1,0}(E) = \dim_{\mathbb{C}} H^{0,1}(E) = g(E) = 1$. The isomorphism above shows that $H^1(E^\circ; \mathbb{C})$ has two one-dimensional Hodge components of types $(1, 0)$ and $(0, 1)$, and $H^1(E^\circ; \mathbb{Q})$ is pure of weight 1.

The radial deformation retraction of $\mathbb{C}^*$ onto S^1 gives

$$H^j(\mathbb{C}^*; \mathbb{Q}) \cong \begin{cases} \mathbb{Q}, & j = 0, 1, \\ 0, & j > 1. \end{cases}$$

Applied to $\mathbb{C}^* = \mathbb{P}^1 \setminus \{0, \infty\}$, [2, Example 2.136] gives an isomorphism of mixed Hodge structures $H^1(\mathbb{C}^*; \mathbb{Q}) \cong \mathbb{Q}(-1)$. Hence this group has Hodge type $(1, 1)$ and is pure of weight 2.

Since each of the $2n - 1$ factors of X_n has no cohomology above degree 1, Deligne's Künneth theorem [5, Proposition 8.2.10] directly gives the following isomorphism of mixed Hodge structures:

$$H^{2n-1}(X_n; \mathbb{Q}) \cong H^1(E^\circ; \mathbb{Q}) \otimes H^1(\mathbb{C}^*; \mathbb{Q})^{\otimes (2n-2)}.$$

Under this isomorphism, Hodge bidegrees add across the tensor factors, so the two summands have types

$$(1, 0) + (2n - 2)(1, 1) = (2n - 1, 2n - 2), \quad (0, 1) + (2n - 2)(1, 1) = (2n - 2, 2n - 1).$$

Both summands are one-dimensional, so the entire group $H^{2n-1}(X_n; \mathbb{Q})$ is pure of weight $4n - 3$. Since $2n < 4n - 3$,

$$W_{2n} H^{2n-1}(X_n; \mathbb{Q}) = 0.$$

□

Proof of Theorem 1.8. Fix $n \geq 2$. The proof of Proposition 5.2 gives

$$W_{2n} H^{2n-1}(Y_n; \mathbb{Q}) = H^{2n-1}(Y_n; \mathbb{Q}), \quad W_{2n} H^{2n-1}(X_n; \mathbb{Q}) = 0.$$

By [4, Theorem 3.2.5(iii)], every regular map $g: X_n \rightarrow Y_n$ satisfies

$$g^*(H^{2n-1}(Y_n; \mathbb{Q})) = g^*(W_{2n} H^{2n-1}(Y_n; \mathbb{Q})) \subseteq W_{2n} H^{2n-1}(X_n; \mathbb{Q}) = 0.$$

Thus every regular map $X_n \rightarrow Y_n$ induces the zero map in degree $2n - 1$.

By Lemma 5.1, there is a continuous map $F_n: X_n \rightarrow Y_n$ with $F_n^* \neq 0$. If Y_n satisfied aBOP, then, since X_n is a smooth connected affine algebraic variety, the definition of aBOP would provide a regular map $g: X_n \rightarrow Y_n$ homotopic to F_n . Homotopy invariance of singular cohomology would then give

$$F_n^* = g^*: H^{2n-1}(Y_n; \mathbb{Q}) \longrightarrow H^{2n-1}(X_n; \mathbb{Q}),$$

which is impossible because $F_n^* \neq 0$ and $g^* = 0$. Therefore Y_n does not satisfy aBOP. Since $n \geq 2$ is arbitrary, the theorem follows. □

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